

Information Gathering Module

1. Target Input

Tester tool mein inmein se koi target de sake:

Domain:

example.com

Subdomain:

app.example.com

Full URL:

https://app.example.com/login

Single IP:

203.0.113.10

Network range:

203.0.113.0/28

2. Main Recon Dashboard

Target add hote hi tester ko ek summary dashboard mile:

Target: example.com

Scope: Authorized

Mode: Passive / Safe Active

Scan Status: Completed

Last Updated: 30 Sep 2026, 03:09 PM

Primary Domain: example.com

Resolved IPs: 4

Subdomains Found: 18

Live Subdomains: 11

Web Applications: 8

Network Hosts: 7

Open Services: 22

URLs Found: 340

API Endpoints: 61

Parameters Found: 88

JavaScript Files: 39

Login Pages: 3

Admin Pages: 2

Upload Features: 1

Payment / Business Pages: 4

High-Priority Targets: 12Dashboard cards clickable honi chahiye. Example:

text

Subdomains Found: 18

Par click karte hi complete subdomain table khul jaaye.

3. Basic Target Profile

Har primary domain/IP ka ek overview profile banana chahiye.

text

Target: example.com

Primary URL: https://example.com

Redirect Chain: http://example.com → https://www.example.com

Page Title: Example Company

HTTP Status: 200

Response Time: 280 ms

Content Type: text/html

HTTPS: Enabled

CDN/WAF: Possible / Detected / Not Detected

Server Header: nginx (if exposed)

IP Address: x.x.x.x

Reverse DNS: Available / Not available

4. Domain and Subdomain Discovery

Tool ko subdomain discovery ka complete output ek table mein dena chahiye.

```
text
example.com
├── www.example.com
├── app.example.com
└── api.example.com
```

Subdomain table

Hostname	IP	Live	HTTP	HTTPS	Title	Scope	Tag
www.example.com	x.x.x.x	Yes	200	Yes	Main website	In scope	Web
api.example.com	x.x.x.x	Yes	401	Yes	API	In scope	API, High
admin.example.com	x.x.x.x	Yes	302	Yes	Login redirect	Review	Admin
dev.example.com	x.x.x.x	Yes					

Filters

```
text
[Live only]
[Dead only]
[API only]
[Admin only]
[Dev / Staging / Test]
[In scope only]
[Needs review]
[No HTTPS]
[HTTP 401 / 403]
[High priority]
```

Automatic priority tags

```
text
api.*      → API / High Priority
admin.*    → Admin / High Priority
dev.*      → Development / Review
staging.*  → Staging / High Priority
test.*     → Testing / Medium Priority
vpn.*      → Remote Access / High Priority
mail.*     → Mail / Network Review
www.*      → Main Web Application
```

5. DNS Intelligence

Tool ko DNS information structured form mein rakhni chahiye.

Host	Record type	Value	Notes
example.com	A	x.x.x.x	Main website IP
www.example.com	CNAME	CDN host	CDN-related
mail.example.com	MX	Mail service	Email infrastructure
example.com	TXT	SPF/verification record	Metadata

Host	Record type	Value	Notes
api.example.com	A	x.x.x.x	API host

6. Network and IP Intelligence

Agar scope mein IP address ya network range approved hai, tool ko host and service intelligence dikhani chahiye.

Network summary

```

text
Authorized Range: 203.0.113.0/28
Total IPs: 16
Live Hosts: 7
Unknown Hosts: 2
Web Hosts: 4
Remote Access Hosts: 2
Database-related Hosts: 1
High-Priority Hosts: 3

```

Host inventory

IP address	Hostname	Status	Role	Related domain	Priority
203.0.113.10	app.example.com	Live	Web server	app.example.com	High
203.0.113.20	api.example.com	Live	API server	api.example.com	High
203.0.113.30	Unknown	Live	Remote access/service	—	Review
203.0.113.40	db.example.com	Live	Database-related	Internal	Critical review

Har host ka profile

```

text
IP Address
Hostname / Reverse DNS
Live Status
Related Domain/Subdomain
Operating System Hint
Device / Vendor Hint
Network Segment
Environment:
- Production
- Staging
- Development
- Unknown

Business Role:

```

- Web server
- API server
- Database
- Mail server
- VPN
- DNS
- Admin/management
- Unknown

Risk Priority
Open Services
Notes

7. Port and Service Inventory

Har authorized live host ke open services ek table mein aa jaayen.

IP	Port	Protocol	Service	State	Version/banner hint	Tag
203.0.113.10	80	TCP	HTTP	Open	Observed	Web
203.0.113.10	443	TCP	HTTPS	Open	Observed	Web
203.0.113.20	22	TCP	SSH	Open	Banner if exposed	Remote admin
203.0.113.30	3389	TCP	RDP	Open	Identified	Remote admin
203.0.113.40	3306	TCP	Database-related	Open	Identified	Database

Service categories

text

Web:

HTTP, HTTPS, HTTP/2, HTTP/3

Remote management:

SSH, RDP, VNC, WinRM

File services:

FTP, SFTP, SMB, NFS

Database services:

MySQL, PostgreSQL, MSSQL, MongoDB, Redis, Elasticsearch

Mail services:

SMTP, IMAP, POP3

Directory / identity:

LDAP, Kerberos, AD-related services

Infrastructure:

DNS, DHCP, SNMP, VPN

Unknown:

Custom/unknown service on port

Service observation output

text

Observation:

Remote administration service is exposed on an authorized host.

Status:

Needs manual review.

Why it matters:

Remote-management services may require access-control and configuration review.

Suggested next phase:

Authorized authentication and exposure review.

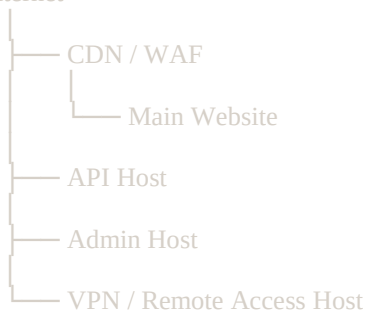
Tool ko **“Port open = vulnerability”** kabhi automatically nahi bolna chahiye.

8. Network Topology View

Tool ke andar visual graph hona chahiye:

text

Internet



Internal/authorized network mein:

text

Gateway



Har node par click karne par:

text

Hostname

IP

Open ports

Running services

Related URLs

Related API endpoints
Technology indicators
Scope status
Criticality
Evidence
Notes

9. Web Application Crawling and URL Map

Tool ko discovered website pages ka tree + table banana chahiye.

```
text
https://app.example.com
|
|— /
|— /login
|— /register
|— /forgot-password
|— /dashboard
|— /profile
|— /settings
|— /orders
|— /orders/{id}
|— /payment
|— /upload
|— /admin
|— /api/v1/
```

URL inventory

URL/path	Method	Status	Auth	Type	Tag	Priority
/login	GET/POST	200	Public	HTML	Authentication	High
/forgot-password	GET/POST	200	Public	HTML	Password reset	High
/dashboard	GET	302	Required	HTML	User dashboard	High
/orders/{id}	GET	200	Required	HTML/JSON	User data	High
/upload	POST	200	Required	HTML/API	File upload	High
/api/v1/users	GET	401	Required	JSON	API	High
/admin	GET	302	Required	HTML	Admin	High

Har URL ke fields

```
text
URL ID
Full URL
Parent URL
HTTP method
HTTP status
```

Content type
Response size
Response time
Authentication required
Test account / role used
Page title
Redirect location
Parameters
Cookies observed
Headers observed
Source of discovery
Screenshot
Notes
Tags
Testing status

10. Endpoint and Parameter Discovery

Tool ko URL milne ke saath uske parameters bhi show karne chahiye.

```
text
/orders?id=1001
/profile?user_id=55
/search?q=mobile
/redirect?next=/dashboard
/download?file=report.pdf
```

Parameter inventory

Endpoint	Parameter	Location	Type hint	Priority tag
/orders	id	Query	Numeric/object ID	Authorization
/profile	user_id	Query	User ID	High
/search	q	Query	Text input	Input handling
/upload	file	Form body	File input	Upload
/login	email	JSON/form body	Email	Authentication
/login	password	JSON/form body	Secret	Authentication

Parameter location types

```
text
URL query parameter
Path parameter
POST form field
JSON body field
XML field
Multipart file field
HTTP header
Cookie
WebSocket message field
GraphQL variable
```

11. API Discovery

Tool ko APIs ko normal URLs se alag section mein properly map karna chahiye.

API sources

text
Crawler
Browser/proxy history
JavaScript files
Swagger/OpenAPI documents
Manual addition
Postman collection import

API inventory

Endpoint	Method	Auth	Inputs	Response type	Tag
/api/v1/login	POST	Public	email, password	JSON	Auth
/api/v1/users/me	GET	Token required	—	JSON	User data
/api/v1/orders/{id}	GET	Token required	id	JSON	Authorization
/api/v1/upload	POST	Token required	file	JSON	File upload

API-specific discovery

text
API base URL
API version
Swagger/OpenAPI availability
GraphQL availability
WebSocket availability
Authentication method
Bearer token usage
Cookie-based session usage
JSON/XML usage
Rate-limit header observations
Error message format
Response schema preview

12. JavaScript Intelligence

Modern web app mein hidden routes aur APIs JavaScript bundles mein milte hain. Tool ko ye central place mein dikhana chahiye.

JavaScript output

text
JavaScript Files Found: 39
External Third-Party Scripts: 8
Potential API Routes: 47
Potential WebSocket URLs: 2
Source Maps: Found / Not found
GraphQL Indicators: Found / Not found
Potential Secret Patterns: 3 needs review

JavaScript table

JS file	Source	Size	Potential routes	Notes
main.js	First-party	1.8 MB	21	API route indicators
vendor.js	First-party	2.3 MB	4	Framework dependencies
analytics.js	Third-party	90 KB	0	Tracking
admin.js	First-party	450 KB	12	Admin route indicators

Tool ko kya extract karna chahiye

text
API endpoints
Relative routes
Absolute URLs
WebSocket endpoints
GraphQL endpoints
External domains
Source map links
Client-side route names
Public configuration endpoints
Potential key/token patterns

Sensitive token/secret output ko always redact karo:

text
Potential secret pattern found:
AIza*****abcd

Status:
Needs manual verification

Full secret default view/report mein show mat karo.

13. Technology Fingerprinting

Tool ko technology information one-screen par deni chahiye.

text
Web server:
nginx — High confidence

Frontend:
React — Medium confidence

Backend:
Possible Node.js — Low/medium confidence

API:
REST JSON — High confidence

CDN:
Cloud-based CDN/WAF indicator — Medium confidence

CMS:
Not detected

Database:
Not confirmed from information-gathering phase

Confidence system

text
High:
Direct header / confirmed response evidence

Medium:
Multiple indirect indicators

Low:
Single weak indicator

Unknown:
No usable evidence

Technology table:

Asset	Technology	Confidence	Evidence
app.example.com	nginx	High	HTTP response header
app.example.com	React	Medium	JS bundle pattern
api.example.com	JSON REST API	High	Response content type
www.example.com	CDN/WAF possible	Medium	Response/network pattern

14. TLS, Headers and Configuration Snapshot

Information gathering stage mein tool ko safe configuration snapshot banana chahiye.

TLS data

text
HTTPS enabled
Certificate subject
Certificate issuer
Certificate start date
Certificate expiry date
Certificate SAN domains
TLS protocol observation
HTTP to HTTPS redirect

Header data

text
Strict-Transport-Security
Content-Security-Policy
X-Frame-Options
X-Content-Type-Options
Referrer-Policy
Permissions-Policy
Cache-Control
Server
X-Powered-By
CORS-related headers

Cookie data

text
Cookie name
Domain
Path
Secure flag
HttpOnly flag
SameSite value
Expiry/session behavior

Output example:

text
Observation:
Cookie “session_id” observed without Secure attribute.

Status:
Configuration observation — needs verification.

Evidence:

HTTP response ID #241.

Next phase:
Session-management review.

15. Authentication Surface Discovery

Information gathering ke time tool auth-related pages identify kare.

text
Login pages found: 3
Signup pages found: 1
Password-reset flows found: 2
MFA/OTP pages found: 1
SSO redirect detected: Yes
Logout endpoint detected: Yes
Admin-login page found: Yes

Authentication map

Feature	URL	Public/Auth	Priority
Login	/login	Public	High
Signup	/register	Public	Medium
Password reset	/forgot-password	Public	High
OTP verification	/verify-otp	Public	High
Logout	/logout	Authenticated	Medium
Admin login	/admin/login	Public	High

16. Important Feature Detection

Tool ko high-value functionality auto-tag karni chahiye:

text
Authentication
Password reset
Admin panel
User profile
Account settings
File upload
Payment
Shopping cart
Orders
User management
API documentation
Webhooks
Search
Download/export

Chat
WebSocket
GraphQL
Cloud storage links

Priority logic

text

High priority:

Admin, API, password reset, user data, payment, upload,
webhook, account settings, export/download

Medium priority:

Search, contact forms, public profiles, dashboards

Low priority:

Static pages, images, favicon, public brochure pages

17. Screenshots and Evidence

Tool ko recon phase mein automatic/manual screenshots save karne ka option dena chahiye.

text

Home page screenshot

Login page screenshot

Admin page screenshot

Password reset screenshot

API documentation screenshot

File upload screenshot

Error page screenshot

Interesting redirect screenshot

Har screenshot ke saath metadata:

text

Screenshot ID

URL

Timestamp

Asset name

HTTP status

Tester name

Tags

Related observation/finding

18. Notes, Tags and Bookmarks

Tester ko har asset, endpoint, port, request aur screenshot par note add karne dena chahiye.

Example:

text

Asset:

api.example.com

Tags:

API, High Priority, Authorization

Note:

Use User A and User B to compare order-data access in next phase.

Status:

Pending authorization testing

Bookmark labels

text

Important

Test later

Need client confirmation

Potential sensitive area

False positive

Out of scope

Evidence saved

19. Search, Filter and Asset Graph

Tool mein global search hona chahiye:

text

Search:

- Domain

- IP

- Port

- URL

- Endpoint

- Parameter

- Technology

- Header

- Cookie

- Tag

- Note

- Screenshot

Example search:

text

Search: admin

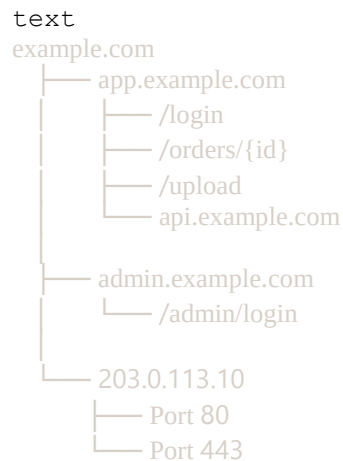
Result:

text

admin.example.com

/admin
/admin/login
admin.js
"Admin Panel" note
Related screenshot

Asset relationship graph



Is graph se tester ko relationship clear dikhega: domain kis IP se related hai, IP par kaunsa service hai, service par kaunsa app/API chal raha hai.

20. Information Gathering Completion Criteria

☐ Information Gathering Completed checkbox tabhi tick ho:

text

- ☒ Primary target reviewed
- ☒ Scope status verified
- ☒ Discovered subdomains reviewed
- ☒ Live hosts identified
- ☒ IP and network-service inventory reviewed
- ☒ Web URLs/endpoints mapped
- ☒ API endpoints mapped
- ☒ Parameters identified
- ☒ Technologies recorded with confidence
- ☒ TLS/header/cookie snapshot saved
- ☒ Login/admin/upload/payment areas tagged
- ☒ High-value targets prioritized
- ☒ Evidence/screenshots saved
- ☒ Notes added for next testing phase
- ☒ Out-of-scope assets excluded

Completion output

Tool end mein tester ko ye summary de:

text

INFORMATION GATHERING COMPLETED

Primary Targets: 2

Subdomains Discovered: 18

Live Web Hosts: 11

Network Hosts: 7

Open Network Services: 22

Web URLs: 340

API Endpoints: 61

Parameters: 88

JavaScript Files: 39

Login Pages: 3

Admin Pages: 2

Upload Features: 1

High-Priority Targets: 12

NEXT TESTING PRIORITY

1. api.example.com

Reason: Authenticated API + object-ID endpoints

2. admin.example.com

Reason: Admin authentication surface

3. /forgot-password

Reason: Account recovery workflow

4. /upload

Reason: File-upload functionality

5. /orders/{id}

Reason: User-specific business data

6. Network host x.x.x.x: remote management service

Reason: Exposed remote-access surface; requires authorized review

Final one-line goal

Information Gathering module ka final output ek complete target map hona chahiye: domains, subdomains, IPs, network hosts, ports/services, URLs, APIs, parameters, technologies, JavaScript routes, authentication pages, high-value features aur next testing priority—sab ek dashboard mein.

Sources

Results for "dekh mujhe bta webpentestar ka kam kya hota he pahle"